Shell cheat sheet

Modern DevOps for Systems Biologists

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Every command here is shown, never run. Type them yourself — a cheat sheet you have not typed is a cheat sheet you cannot use under pressure.

Where am I

Command	What it does
`pwd`	print the working directory
`ls -la`	list everything, including dotfiles, with permissions
`cd -`	go back to the directory you were just in

TODO: add the two or three you actually use every day.

Moving things around

```
cp -r source/ destination/
mv old_name new_name
rm -i unwanted.txt
```

TODO: a sentence on why `rm` has no undo, and what to do instead.

Command	What it does

Looking inside files

Command	What it does
`less file.txt`	page through a file; `q` quits
`head -20 file.csv`	the first twenty lines
`wc -l file.csv`	count the lines
`grep -n "pattern" file.py`	find a pattern, with line numbers

TODO

Pipes and redirection

```
ls *.csv | wc -l
python analysis.py > results.txt 2> errors.txt
grep "ERROR" log.txt | head
```

TODO: the one idea — a program reads a stream and writes a stream, and the shell is what connects them.

Talking to another machine

```
ssh user@hpc3.rcic.uci.edu
scp results.csv user@hpc3.rcic.uci.edu:~/project/
rsync -av --progress local_dir/ user@hpc3.rcic.uci.edu:~/project/
```

TODO: when to reach for `rsync` instead of `scp`.

On the cluster

```
squeue -u $USER
sbatch job.sh
scancel JOBID
```

TODO: the minimum viable batch script.

Getting yourself unstuck

Situation	What to type
a command is running away	`Ctrl-C`
you want the manual	`man command`, or `command --help`
you do not know where a program is	`which command`

TODO